

Multi-Tenant SaaS POS System

**A SYNOPSIS OF MAJOR PROJECT
for the Degree of**

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE &**

**ENGINEERING
SUBMITTED TO**



**LNCT UNIVERSITY
Bhopal (M.P.)
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**Year - 3rd semester - 6th
Session: JAN-JUNE 2026**

Synopsis

Project Title

Multi-Tenant SaaS POS System

1. Problem Statement

Retail businesses such as malls, supermarkets, and retail chains often face difficulties in managing sales, inventory, employees, customers, and payments across multiple branches using traditional POS systems. Most existing POS solutions are either hardware-dependent, costly, lack scalability, or do not support real-time analytics and centralized control.

Small and medium enterprises also struggle with POS systems that do not provide role-based access, subscription models, or cloud-based multi-tenant architecture. This creates inefficiencies in business operations, inventory mismatches, limited reporting capabilities, and poor customer experience.

Hence, there is a strong need for a modern, cloud-based, scalable, secure, and enterprise-ready SaaS POS system that can efficiently handle multi-store operations and real-time business analytics.

2. Objective of the Project

The primary objective of this project is to design and develop a production-ready enterprise SaaS POS system that automates retail operations using modern full-stack technologies.

Specific Objectives:

- To develop a multi-tenant POS platform for multiple stores and branches
- To provide role-based dashboards for Cashier, Branch Manager, and Store Admin
- To manage sales, inventory, refunds, customers, and employees
- To implement secure authentication and authorization
- To support multiple payment gateways
- To generate real-time reports and analytics
- To create a scalable SaaS subscription model

3. Proposed Solution

The proposed solution is a web-based SaaS Point of Sale system built using React for frontend, Spring Boot for backend, and MySQL as the database.

The system follows a Client–Server Architecture where:

- The frontend provides interactive dashboards and cashier interfaces
- The backend handles business logic, security, and API processing
- The database stores transactional and master data securely

The solution supports:

- Multi-branch operations
- Centralized inventory tracking
- Secure payment processing
- Business intelligence dashboards
- Subscription-based SaaS model

5. Scope of the Project

The scope of the project includes:

- Retail shops, supermarkets, malls, and franchises
- Multi-store and multi-branch management
- Inventory and stock control
- Order and refund processing
- Customer loyalty management
- Employee and role management
- Real-time analytics and reporting
- SaaS subscription plans

The project does not include hardware integration such as POS machines or barcode scanners (can be added in future).

6. System Requirements

Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB
- Storage: 256 GB SSD or higher
- Internet Connection: Required

Software Requirements

- Operating System: Windows / Linux / macOS
- Frontend Runtime: Node.js (v18+)
- Backend Runtime: Java JDK 17+
- Database Server: MySQL 8+
- IDE: VS Code / IntelliJ IDEA
- Browser: Google Chrome / Firefox

6. Technology Stack

Frontend

- React.js (Component-based UI)
- Redux Toolkit (State Management)
- shadcn/ui (UI Components)
- Axios (API Communication)
- Tailwind CSS (Styling)
- Cloudinary (Image Management)

Backend

- Java Spring Boot
- Spring Security (JWT Authentication)
- JPA / Hibernate (ORM)
- RESTful API Architecture

Database

- MySQL (Relational Database)

Tools & Frameworks

- Git & GitHub (Version Control)
- Postman (API Testing)
- Maven (Dependency Management)
- Razorpay & Stripe (Payment Gateways)

7. System Features

7.1 Cashier Terminal Interface

- Product search with barcode scanning (F1 hotkey)
- Smart shopping cart with quantity management
- Multiple payment methods (Card, Cash, UPI)
- Customer selection & management
- Discount system (percentage or fixed amount)
- Order notes & special instructions
- Refund processing
- Complete order history with date filters
- Invoice generation & printing
- Shift reports with payment breakdown

7.4 Branch Manager Dashboard

- Order management with search & filters
- Real-time inventory tracking
- Low stock alerts with product details
- Employee management (active/inactive status)
- Branch analytics & performance metrics
- Top selling products tracking
- Branch settings & configuration
- Payment method breakdown
- Intelligent business alerts

7.3 Store Admin Panel

- Multi-location management (unlimited stores & branches)
- Enterprise-wide analytics dashboard
- Employee management across all locations
- Product catalog with categories
- Customer loyalty points system
- Sales trends & growth tracking
- Subscription plan management
- KPI tracking (Total sales, Average order value, etc.)

7.4 Business Intelligence Features

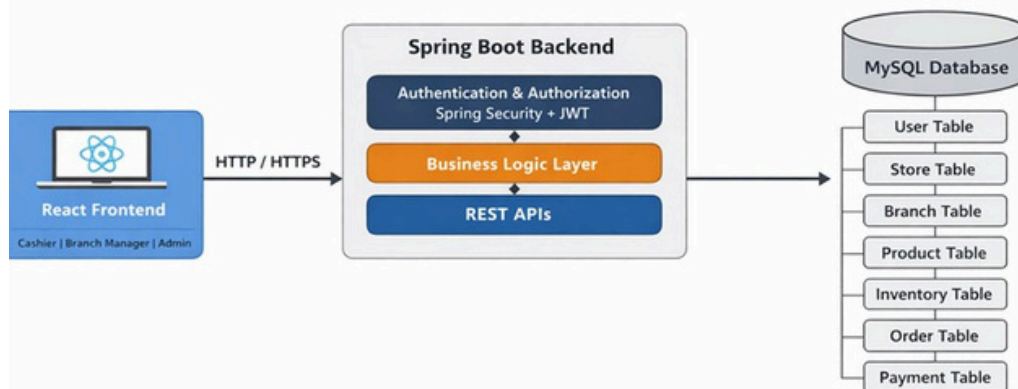
- Daily sales trends (Last 7 days)
- Payment method distribution charts
- Branch performance comparison
- Customer purchase history
- Sales growth percentage tracking
- Average order value analytics
- Top products & categories reports

8. Proposed Methodology

The project follows a modular and incremental development methodology:

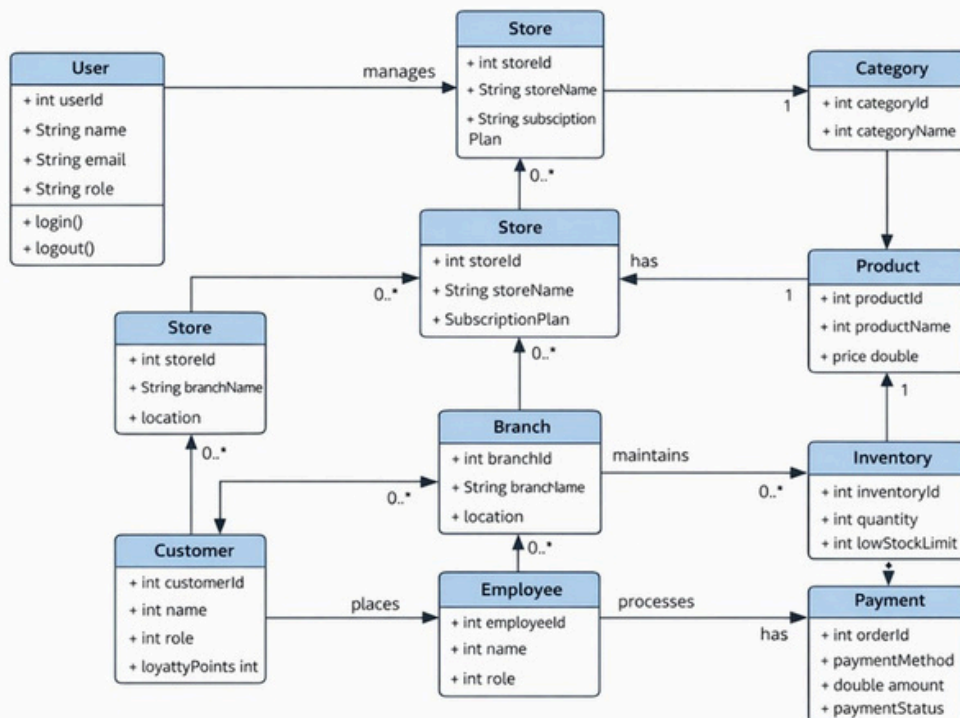
- Requirement Analysis
- System Architecture Design
- Database Design (ER Diagram)
- Backend API Development
- Authentication & Security Implementation
- Frontend UI Development
- Redux State Integration
- Payment Gateway Integration
- System Testing
- Deployment Preparation

• System Architecture Diagram



(Client → API → Database Architecture)

• ER Diagram / Database Design



9. Expected Outcome

- Fully functional enterprise SaaS POS system
- Secure and scalable architecture
- Real-time inventory and sales tracking
- Professional dashboards for different user roles
- Portfolio-ready production-grade project
- Improved understanding of enterprise application development

10. Future Scope

- Mobile POS application (Android/iOS)
- Barcode & QR code scanner integration
- AI-based sales forecasting
- GST and tax automation
- Cloud deployment (AWS/GCP)
- Multi-language support
- Offline mode support